UNIVERSIDADE DE SÃO PAULO

Instituto de Ciências Matemáticas e de Computação

RESEARCH PROJECT

Ph.D.

Asymptotic for Partition Functions of External Field Models in Random Matrix

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Resumo

A Teoria de Matrizes Aleatórias demonstra aplicabilidade em uma gama diversa de áreas na matemática e na física, com destaque no estudo de mecânica estatística e probabilidade. No estudo da distribuição de autovalores de matrizes aleatórias, analogias à sistemas físicos - especialmente termodinâmicos - se tornam evidentes e motivações físicas podem ser consideradas. A função de normalização da distribuição de tais autovalores, por exemplo, é tipicamente identificada com a infame função partição, que em termodinâmica, por exemplo, codifica informações sobre propriedades do sistema descrito. Desenvolvimentos recentes tem progredido intensamente na compreensão da expansão assintótica, no limite termodinâmico, da função partição de sistemas termodinâmicos equivalentes à ensembles de matrizes aleatórias de uma classe determinada invariante. O objetivo deste trabalho é aplicar tais desenvolvimentos em um sistema de difícil tratamento denominado modelo de matrizes aleatórias com campo externo. Sua relevância é principalmente expressa pelo comportamento de Kernel do tipo Pearcing apresentado por tal modelo, o tornando atraente para melhor entendimento - e extensão - da corrente teoria de matrizes aleatórias.

Abstract

The theory of Random Matrices finds its way in a diverse range of areas of Mathematics and Physics, and in particular to statistical mechanics. In the study of the spectra of eigenvalues of some random matrices ensembles, a physical analogy regarding thermodynamic systems becomes evident and the use of physical motivations in its study is of great importance. An example of such motivation is the study of the partition function, which holds in thermodynamics a great deal of information on the system at hand. Recent developments have been intensely unraveling new critical phenomena in random matrix theory, through a deep understanding of asymptotic expansions of the associated partition functions in the thermodynamic limit. The main goal of this work is to apply these developments in a particularly difficult situation denominated external field random matrix model. Its relevant is well assured by its Pearcing edge uncommon scaling - making it vary appealing not only for the better understanding of the current theory but for extending it for some more general cases.

This proposal was elaborated according to FAPESP norms, following the guidelines available at `https://fapesp.br/253/projeto-de-pesquisa`

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1 Introduction

Random Matrices can be though of in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. There are many instances of "models" of random matrices, each one with its symmetries and own probability density. Those are what we call *ensembles*. The definition of ensembles is of great use when modeling phenomena with the use of random matrices. Each ensemble corresponds to particular symmetries and scaling properties designed to proper describe the chosen system. More concretely, in the context of Random Matrix Theory (RMT) we have the following definition.

Definition 1.1. *A random matrix ensemble $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ is a set of matrices $\mathcal{S}$ of fixed size equipped with a p.d.f. $\mathcal{P} : \mathcal{S} \to [0, \infty)$. If we say that a matrix $\hat{X}$ is drawn from the random ensemble $\mathcal{E}$, we are saying that it is a random variable assuming values in $\mathcal{S}$ with p.d.f. $\mathcal{P}$.*

Perhaps the first instance of the modeling by random matrices was in the works of Wigner on the scattering of heavy nuclei where a consensus emerged; the theory of resonances may be statistical. Wigner proposed, and subsequently this was formalized as the Wigner-Dyson-Mehta (WDM) conjecture, that the curve of resonances distribution must hold some universality that only depends on a few of the main properties of the nuclei - namely its symmetries. This modeling was repeated for many other highly correlated processes and was very successful as a general theory of random correlated systems. This can now be observed in the platitude of posterior system modeled with such statistics, for example: Ulam's Problem (ULAM, 1961), Zeros of the Riemann Zeta Function (GOLDSTON; MONTGOMERY,), Matching Random Sub-sequences (MAJUMDAR, 2007), Tiling problems (Aztec Diamond, for instance) and Interacting Random Walks, to name a few.

Interestingly, the models we will deal with can be interpret as simple Point Processes, a random point dynamic with no intersecting points. Especially, they can be described as Determinantal Point Processes, that is, their distribution has specific form that is better expresses in the following definitions.

Definition 1.2. *(Point Processes) Let Λ be a complete separable metric space. A* ***point process*** *$\mathbb{P}$ on Λ is a probability measure on the space of all configuration of points χ or counting measures $\mathcal{N}(\Lambda)$. We say $\mathbb{P}$ to be a n-point process if $\mathbb{P}(\#\chi = n) = 1$ or, equivalently, $\mathbb{P}(\xi(\Lambda) = n) = 1$. We say that the counting measure ξ is simple if $\xi(\{x\}) \leq 1$ for all $x \in \Lambda$, then, the point process is simple if $\mathbb{P}(\xi$ is simple$) = 1$.*

Definition 1.3. *Consider a point process P on a complete separable metric space Λ, with reference measure $\mathtt{p}_N$, all of whose correlation functions ρ_k exist. If there is a function $K : \Lambda \times \Lambda \to \mathbb{C}$ such that*

$$\rho_k(x_1, x_2, \ldots, x_k) = \det(K(x_i, x_j))_{i,j=1}^k \tag{1.1}$$

for all $x_1, x_2, \ldots, x_k \in \Lambda$, $k \geq 1$, then we say that P is a ***determinantal point process****. We call K the* ***correlation kernel*** *of the process.*

The so called Kernels K will play a major role in the description that follows. Typically these model shave some size parameter N: In Coulomb Gases it would indicate the number of particles in the system, for example. One is usually interested in the existence of limiting distributions as the size parameter increases to infinity. (TRACY; WIDOM, 2006) Universality theorems in this context are statements that some operator K is the limit for a class of operator K_n. For example, There is a universal law in the edge of the distribution of the eigenvalues. Already on Wigner's Matrices one can notice an instance of these distributions: The Tracy-Widom (TRACY; WIDOM, 1993), defined by, given some parameters,

$$\lim_{n\to\infty} \mathtt{p}\left(\frac{\sum_{k=1}^{n}\left(\lambda_k - z_n^{(\beta)}\right)}{s_n^{(\beta)}} \leq t\right) = \mathbb{F}_\beta(t), \tag{1.2}$$

describes the distribution of the largest eigenvalue, a distribution highly influenced by two opposing forces, the eigenvalue repulsion and the potential contention. The emerging distribution should be viewed as the random matrix analogue of the central limit theorem and the normal distribution. The Airy functions (related to the Airy Kernel and Tracy-Widom function) themselves are given by integrals in which the exponents have a cubic singularity, arising from the coalescence of two saddle points in an asymptotic analysis. As is true for many kernels $K_n(x, y)$, the kernel here can be expressed by

$$\int \int f(s,t)\, \mathrm{e}^{\phi_n(t,s;x,y)}\, \mathrm{d}s\mathrm{d}t. \tag{1.3}$$

Especially, for this case, the coalescence of two saddle points leads to the fold singularity $\phi_2 = 1/3z^3 + \lambda z$.

However, in the scope of this work, we will be concerned with the *Pearcey Processes*. The Pearcey process describes the local eigenvalue statistics of large random matrices near the points of the spectrum where the limiting mean eigenvalue density vanishes as a cubic root. Pearcey functions are given by integrals in which the exponents

$$\phi_3(x) = \frac{1}{4}z^4 + \lambda_2 z^2 + \lambda_1 z \tag{1.4}$$

have a quartic singularity, arising from the coalescence of three saddle points - these are the functions which give rise to the Pearcey Processes. This is the so called Pearcey Kernel. Its theory emerged on the study on the level spacing distribution for random Hermitian matrices in an external field. More precisely, let $\hat{H}$ be an $n \times n$ GUE matrix (with n even), suitably scaled, and $\hat{H}_0$ a fixed Hermitian matrix with eigenvalues $\pm a$ each of multiplicity $n/2$. Let $n \to \infty$. If a is small the density of eigenvalues is supported in the limit on a single interval. If a is large then it is supported on two intervals. There must be some point where the gap closes, at this point, the limiting eigenvalue distribution is described by the Pearcey kernel. These models of matrices have p.d.f.

given by functions

$$\mathcal{P}(M)\mathrm{d}M = \frac{1}{Z_{N,\beta}}\,\mathrm{e}^{-N(\operatorname{Tr} V(M)+AM)}\,\mathrm{d}M, \quad M \in \mathbb{S} \quad \text{and} \quad A \in \Lambda, \tag{1.5}$$

where $Z_{N,\beta}$ is a partition function, such that $\mathcal{P}$ is a probability density of the matrices. The potential $V : \mathbb{R} \to \mathbb{R}$ is a suitably regular function, acting as a confining potential on the eigenvalues $\lambda_1, \ldots, \lambda_N$. The set Λ is the set of diagonal real matrices (generally hermitian, but simply diagonal) and $\mathbb{S}$ is such as the real, complex, or quaternionic sets. Especially, we can describe the model by its partition function

$$Z_{N,\beta} = \int \mathrm{e}^{-N(\operatorname{Tr} V(M)+AM)}\,\mathrm{d}M. \tag{1.6}$$

Moreover, we say that $A = \operatorname{diag}(a_1, \cdots, a_p)$ and that each a_i has multiplicity n_p. Note that $\sum n_p = n$. This type of model has many reasons to exist and possible interpretations. One could think of a system with impurities where the Hamiltonian looked like $\mathcal{H} = \mathcal{H}_0 + V$ where V is a potential for the impurity in the system and $\mathcal{H}_0$ is some deterministic matrix. With that, one can create critical points of some new type and study its properties by the asymptotic of its partition function.

1.1 Orthogonal Polynomial Ensembles

The subclass of invariant matrix ensembles this section refers to are those self-adjoint matrices $\hat{X}$ with p.d.f.s $\mathcal{P}(\hat{X})$ such that one has the decomposition $\mathcal{P}(\hat{X}) = \prod_{i=1}^{n} w(\lambda_i)$, where $\{\lambda_i\}$ are the undistinguishable eigenvalues of $\hat{X}$ and $w(\lambda)$ is some continuous weight function that is real valued on $\mathbb{R}$ and decays fast enough as $|\lambda| \to \infty$. This is the same as removing the constrain on the weight function of the classical ensembles.

Definition 1.4. *Let $\beta = 1, 2, 4$, fix $n \in \mathbb{N}$ and let $w(\lambda)$ be a continuous, non-negative,* *real-valued function such that* $w(\lambda) = o(\lambda^{-\beta(n-1)-1})$*. Then, the j.p.d.f.*

$$\mathtt{p}^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta) = \frac{1}{\mathcal{N}_{n,\beta}^{(w)}} \prod_{i=1}^{n} w(\lambda_i) |\Delta_n(\lambda)|^\beta \tag{1.7}$$

describes the n-sized orthogonal polynomial ensemble (OPE) when $\beta = 2$*.*

The condition on the weight function guarantees that the j.p.d.f. has a well-defined normalization constant. If one wishes to integrate the determinantal structures given by the Jacobian against the product of weight functions, it is beneficial to use monic polynomials $\{q_j(\lambda)\}_{j=0}^{\infty}$ that are skew-orthogonal with respect to the weight $w(\lambda)$. We focus on the results based on $\beta = 2$ but they all have natural extensions for $\beta = 1, 4$, even if they are not to focus here.

Definition 1.5. *With* $w(\lambda)$ *a suitable function, define the following inner product on polynomial space:*

$$\langle f, g \rangle^{(2)} := \int_{\mathbb{R}} f(x) g(x) w(x) \mathrm{d}x \tag{1.8}$$

For $\beta = 2$*, let* $\{q_j^{(\beta)}\}_{j=0}^{\infty}$ *be a family of monic polynomials such that each* $q_j^{(\beta)}(\lambda)$ *is of degree* j*. Impose also that exists finite positive constants* $\{r_j^{(\beta)}\}_{j=0}^{\infty}$ *s.t.* $\langle q_j^{(2)}, q_k^{(2)} \rangle^{(2)} = r_j^{(2)} \chi_{j=k}$*. Then, we say that the family* $\{q_j^{(2)}(\lambda)\}_{j=0}^{\infty}$ *is orthogonal in respect to* $w(\lambda)$*.*

We can now express the eigenvalue j.p.d.f. $\mathtt{p}^{(w)}$ as a determinantal process

Theorem 1.6. *Let* $\{q_j^{(\beta)}(x)\}_{j=0}^{\infty}$ *be as in the definition above, and let* $n \in \mathbb{N}$*. Then, the kernel corresponding to the complex case,*

$$K_n^{(2)}(x, y) := \sqrt{w(x) w(y)} \sum_{k=0}^{N-1} \frac{1}{r_k^{(2)}} q_k^{(2)}(x) q_k^{(2)}(y). \tag{1.9}$$

Then, the eigenvalue j.p.d.f. is given by $p^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta) = \det \left[K_n^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^{n}$,

The significance of this theorem is that, for $\beta = 2$ -as would be analogue for

$\beta = 1, 4$ - the kernel $K_n^{(\beta)}(x, y)$ have *Reproducing properties.* Especially, for $\beta = 2$,

$$\int_{\mathbb{R}} \det \left[K_n^{(2)}(\lambda_i, \lambda_j)\right]_{i,j=1}^{k+1} \mathrm{d}\lambda_{k+1} = (n-k) \det \left[K_n^{(2)}(\lambda_i, \lambda_j)\right]_{i,j=1}^{k} \tag{1.10}$$

Thus, one may integrate the j.p.d.f. over $\mathrm{d}\lambda_{k+1} \cdots \mathrm{d}\lambda_n$ to obtain the *k-point correlation function*

$$\mathtt{p}_k^{(w)}(\lambda_1, \lambda_2, \ldots, \lambda_k; n; \beta) := \frac{n!}{(n-k)!} \int_{\mathbb{R}^{n-k}} \mathtt{p}^{(w)}(\lambda_1, \lambda_2, \ldots, \lambda_n; \beta) \mathrm{d}\lambda_{k+1} \cdots \mathrm{d}\lambda_n \tag{1.11}$$

$$= \det \left[K_n^{(2)}(\lambda_i, \lambda_j)\right]_{i,j=1}^{k} \tag{1.12}$$

Therefore the eigenvalue density $p^{(w)}$ is given by the formula $p^{(w)}(\lambda; \beta) = K_n^{(2)}(\lambda, \lambda)$. All that remains to make $\mathtt{p}^{(w)}(\lambda)$ fully explicit is knowledge of the orthogonal polynomials $\{q_j^{(2)}(\lambda)\}_{j=0}^{\infty}$. In the Gaussian, Laguerre and Jacobi cases, there are Hermite, Laguerre and Jacobi orthogonal polynomials, respectively.

1.2 Partition Functions in Random Matrix

In the research proposed here, we will focus on the study of *normal random matrices*, which consists of the space of normal matrices, equipped with a proper probability distribution. In the beginning of its book, Feynman states (FEYNMAN, 1998) that the key principle of the statistical mechanics is that the probability that a system attains a state with energy E, in equilibrium, is given by a function $\frac{1}{Z} \mathrm{e}^{\frac{-E}{kT}}$ where Z is the partition function. In a more general sense there is a relation between the partition function and the free energy of the system, that, in itself, relates to the entropy. Knowing well the partition function is a way to describe with great detail the macro-properties of such a system.

There is a great effort in the community of random matrix theory to study expansions of the partition function of various classes of Random Matrix. Recently, some advance has been made in the works of Sug-Soo Byun et al. (BYUN; KANG; SEO, 2023). The work they done was on Random Normal Matrix and Planar Ensembles (with Determinantal and Pfaffian structures, respectively), which present partition functions

$$Z_N^{(\beta)} := \int_{\mathbb{C}^N} \prod_{j>k=1}^{N} |z_j - z_k|^\beta \prod_{j=1}^{N} \mathrm{e}^{-\frac{\beta N}{2} Q(z_j)}\, \mathrm{d}A(z_j) \tag{1.13}$$

where N is the size of the matrix, β is relative to the generic entries dimension and $\mathrm{d}A(z) := \mathrm{d}^2 z/\pi$ is the area measure. Here, $Q : \mathbb{C} \to \mathbb{R}$ is called the confining potential and satisfies some suitable conditions. More over one can show for invariant ensembles that the partition function relates directly to the norming constants of the pertinent orthogonal polynomials for the ensemble. This is expressed precisely as the proposition.

Proposition 1.7. *When dealing with a p.d.f. of invariant measure, one can write*

$$\mathcal{Z}_{n,2} = n! \prod_{k=0}^{n-1} r_k^{-1}, \tag{1.14}$$

where r_k is the normalizing constant as in $\langle q_j^{(2)}, q_k^{(2)} \rangle^{(2)} = r_j^{(2)} \chi_{j=k}$.

Then, using for example the integral representation of orthogonal polynomials, one can get good asymptotic for the partition function and consequently for the free energy of a given system. In this vein, using orthogonal polynomials, integral asymptotics by Laplace method, and quadrature methods, in (BYUN; KANG; SEO, 2023) they derived large-N asymptotic expansions up to the $O(1)$-terms for $Z_{N,\beta}$ when V is a radially symmetric potential, in the form

$$Z_{N,\beta=2} = -I^V N^2 + \frac{1}{2} N \log N + E^V N + G \log N + F^V + o(1), \quad N \to \infty. \tag{1.15}$$

In such expansion, for a large class of potentials V the first term I^Q was known for a long time, being given by the energy of the associated equilibrium measure. The coefficient 1/2 of the order $N \log N$, as well as the entropy term E^V, were known from recent work of Leblé and Serfaty (LEBLé; SERFATY, 2017), also for rather large classes of V. The remaining terms in this expansion are new, and they are computed exploring the radially symmetry imposed on the potential. Strikingly, the term G appears to be universal in V, depending solely on the connectivity of the limiting spectrum of eigenvalues. More precisely, they noticed that G depends on whether the limiting spectrum is an annulus or a disc which, surprisingly, seems to not have been observed in the vast previous literature on the matter.

As was said, the study of the partition function is a major way to describe thermodynamic systems. With these new developments it is expected that many systems could be better understood and described by the asymptotic given. In that way this poses a great opportunity for development in the field. More than that, by the great connection that the field of random matrices holds with many other fields is possible that by the study of this work some suggestions on the behavior of many other related problems become plausible. Especially, we will be interested in external field model.

1.3 External Field Models

Consider now that we couple a deterministic source matrix $\hat{A}$ to the random matrix $\hat{M}$, in a way that the weight

$$\mathrm{d}\mu_n(M) = \frac{1}{Z_A} \exp\left\{-n\left(\mathrm{Tr}\left(V(\hat{M}) - \hat{A}\hat{M}\right)\right)\right\}\mathrm{d}M. \tag{1.16}$$

describes the model distribution. We take $\hat{A}$ to be an Hermitian matrix and $\mathrm{d}M$ to be the entry-wise Lebesgue measure. The eigenvalues of $\hat{M}$ represent the energy levels of a system without time-reversal invariance. When the external source $\hat{A}$ is nonzero and $V(\hat{M}) = \hat{M}^2/2$, this measure arises in the study of Hamiltonian that can be written as the sum of a random matrix and a deterministic source matrix. In this definition,

$$Z_A = \int \exp\left\{-n\left(\mathrm{Tr}\left(V(\hat{M}) - \hat{A}\hat{M}\right)\right)\right\}\mathrm{d}M \tag{1.17}$$

Without any loss of generality, we can assume $\hat{A}$ to be diagonal.

Of course, when $\hat{A} = 0$, that is, no external source, and $V(\hat{M}) = \hat{M}^2/2$, this measure describes the Gaussian Unitary Ensemble - GUE. For reasonable choices of $V(z)$ the spectrum tends to accumulate on closed intervals of the real axis. Introducing an external source A can have the effect of perturbing the expected position of the spectrum. Some study has been done on the case for the matrix $\hat{A}$ with two eigenvalues $\pm a$, each of multiplicity $n/2$. When a is sufficiently small, the eigenvalues of $\hat{M}$ accumulate with probability one on a single interval, just as when $\hat{A} = 0$. As a increases the interval splits into two disconnected intervals. There is a transitional value of a between these two cases where the local eigenvalue density is described by the so-called *Pearcey process*. This is precisely the case witch we will explore an asymptotic for. Define the monic polynomial

$$P_n(z) = \int \det(z - \hat{M}) \mathrm{d}\mu_n(M) \tag{1.18}$$

then, we have the following proposition

Proposition 1.8. *Suppose $\hat{A}$ has p distinct eigenvalues a_i with respective multiplicity n_i, so that $n_1 + \cdots + n_p = n$. Let $n^{(i)} = n_1 + \cdots + n_i$ and $n^{(0)} = 0$. Define*

$$w_j(x) = x^{d_j - 1} \, \mathrm{e}^{-(V(x) - a_j x)}, \quad j = 1, \cdots, n. \tag{1.19}$$

where $i = i_j$ is such that $n^{(i-1)} < j < n^{(i)}$ and $d_j = j - n^{(i-1)}$. Then the following hold

- *There is a constant $\tilde{Z}_n$, with $\mathrm{d}\lambda = \mathrm{d}\lambda_1 \cdots \mathrm{d}\lambda_n$, such that*

$$P_n(z) = \frac{1}{\tilde{Z}} \int \prod_{j=1}^{n} (z - \lambda_j) \prod_{j=1}^{n} w_j(\lambda_j) \prod_{i>j} (\lambda_i - \lambda_j) \mathrm{d}\lambda \tag{1.20}$$

- *Let*

$$m_{j,k} = \int_{-\infty}^{\infty} x^k w_j(x) \mathrm{d}x \tag{1.21}$$

Then we have the determinantal formula

$$P_n(z) = \frac{1}{\tilde{Z}_n} = \begin{vmatrix} m_{1,0} & m_{1,1} & \cdots & m_{1,n} \\ \vdots & \vdots & \ddots & \cdots \\ m_{n,0} & m_{n,1} & \cdots & m_{n,n} \\ 1 & z & \cdots & z^n \end{vmatrix} \tag{1.22}$$

- *For* $j = 1, \cdots, p$,

$$\int_{-\infty}^{\infty} P_n(x) x^j \, \mathrm{e}^{-(V(x)-a_j x)} \, \mathrm{d}x = 0, \quad j = 0, \cdots, n_i - 1 \tag{1.23}$$

and these equations uniquely determine $P_n(x)$.

The relations can be viewed as multiple orthogonality conditions for the polynomial P_n. There are p weights $\mathrm{e}^{-(V(x)-a_j x)}$, $j = 1, \cdots, p$, and for each weight there are a number of orthogonality conditions, so that the total number of them is n. This point of view is especially useful in the case when $\hat{A}$ has only a small number of distinct eigenvalues. Let $\sum_n$ be the collection of functions $\sum_n := \{x_j \, \mathrm{e}^{a_i x} \, | i = 1, \cdots, p, \; j = 0, \cdots, n_i - 1\}$ We start with a lemma.

Lema 1.9. *There exists a unique function* Q_{n-1} *in the linear span of* $\sum_n$ *such that*

$$\int_{-\infty}^{\infty} x^j Q_{n-1}(x) \, \mathrm{e}^{-V(x)} \, \mathrm{d}x = 0, \quad j = 0, \quad , n-2, \tag{1.24}$$

and

$$\int_{-\infty}^{\infty} x^{n-1} Q_{n-1}(x) \, \mathrm{e}^{-V(x)} \, \mathrm{d}x = 1. \tag{1.25}$$

Consider $P_0, \cdots, P_n$, a sequence of multiple orthogonal polynomials such that $\deg P_k = k$, with an increasing sequence of the multiplicity vectors, so that $k_i \leq l_i, i = 1, \cdots, p$, when $k \leq l$. Consider the biorthogonal system of functions, $\{Q_k(x), k =$

$0, \cdots, n-1\}$,

$$\int_{-\infty}^{\infty} P_j(x) Q_k(x)\, \mathrm{e}^{-V(x)}\, \mathrm{d}x = \delta_{jk}, \quad \text{for} j, k = 0, \cdots, n-1. \tag{1.26}$$

Define the kernel $K_n(x,y) = \mathrm{e}^{-\frac{1}{2}(V(x)+V(y))} \sum_{k=0}^{n-1} P_k(x) Q_k(y)$. This kernel, as in the simplest case in (1.12), will enable us to determine asymptotic on the equilibrium measure.

2 Justification

The student has its Bachelor in Computational Physics and is currently finishing his Master in Mathematics. As an undergrad, he has worked on the general concepts of Random Matrix and has touched on some important models by the implementation of a simulation method for the eigenvalues equilibrium measures. He also ventured in some works on probability and published two articles in the general field: *Simulating simple random walks with a deck of cards* (DOI: 10.1214/25-BJPS635) and *A Central Limit Theorem for intransitive dice* (DOI: 10.30757/ALEA.v22-46). As a master student, he has worked on partition functions and explicitly studied the work done by Sug-Soo on the asymptotic of normal and planar ensembles. In this manner, it is of our understanding that this Ph.D. research would be a natural and important step in the construction of experience and expertise in the field. Moreover, we also think the student is a good candidate for such project.

3 Objectives

Here, we discuss the general scientific objectives of the project. Ultimately, as was discussed before, we want to be able to express the asymptotic of the partition function for some model of the external field ensemble with as much precision as possible. Let's lay out the path of this project to obtain such a goal.

Our main object is the partition function $\mathcal{Z}$ and at the same time, the free energy $\ln(\mathcal{Z})$. If we think of models described so far, we can see that generally it is possible to express such a function in terms of the associated norming constants, i.e., in terms of the orthogonal polynomial associated with the problem and consequently, because of the reproducing property, the norming constant. For this work we will deal with what was called the bi-orthogonal ensemble. Naturally because of the different mathematical structure, the partition function won't have the same expression as before in terms of norming constants. It is expected however to be some analogous formula. Our first goal is then to find an expression of the partition function in terms of these norming constants for the bi-orthogonal polynomials that define the problem.

> Express the partition function in terms of the norming constants for the bi-orthogonal polynomials that define the problem of the bi-orthogonal ensemble

Interestingly, such polynomials have an accessible asymptotic. This was explored and determined in a recent work (BLEHER; KUIJLAARS, 2006); Pavel and Arno have found the asymptotic for the large n limit of Gaussian random matrices with external source, in terms of the associated Riemann-Hilbert problem for the model. So, for this step, the objective is to translate the results into the asymptotic of the norming constants and plug it into our know partition function formula developed in (BYUN; KANG; SEO, 2023). Finally, possessing the asymptotic for the bi-orthogonal polynomials we should be able with the aid of standard techniques to express the asymptotic for $\mathcal{Z}$. This is therefore a very natural final objective.

> Connect the asymptotic given by the Riemann-Hilbert problem by Pavel and Arno into the asymptotic for the bi-orthogonal polynomials using the formula in (BYUN; KANG; SEO, 2023) to get out asymptotic for the partition function.

Possessing such asymptotic we can validate and check what kind of gains or loses we had in applying the method described in (BYUN; KANG; SEO, 2023) and studied in the student's masters. Therefore checking for the usefulness of such technique in this relatively new model.

4 Work Plan and Schedule

In this section we describe which activities we plan on achieving during the time of execution of the scholarship. We also display the timetable for when we expect them to be done and the given priorities for each time period.

4.1 Planned Activities

1. Bibliographical Research: Praxis in Mathematics, the bibliographic research is expected and needed as a way to get used to important concepts related to the field of research;

2. Guidance Meetings: As usual for a Ph.D., guidance meetings with the supervisor are both a way of developing the necessary tools of the field and validating the work done so far in the research;

3. Program's Courses: Expected from any student enrolled in the Mathematics program, there is a quantity of needed course credits one must fulfill;

4. Qualification Exam: A demand from the program, the qualification exam marks the passage to a Ph.D. candidate;

5. FAPESP Technical Reports: The technical reports are expected to be completed every year in accordance to the FAPESP guide;

6. Research Problem Preliminary Study: The first step in the research, along with the bibliographic research, this step marks the beginning of taking the research problem face on;

7. Research Problem Solution Possibilities: Scavenging the bibliography pops up many possibilities of solutions to the problem, here we endorse some of the possibilities;

8. Implementation of Possible Solutions: Of course, this step is self-explanatory: One must apply the solutions one has researched;

9. Results Formulation and Publications: This is one of the final steps in the research process. Formulating the results and publishing are necessary steps for the completion of the Ph.D.;

10. Thesis Development: The final product of the Ph.D. is, invariably, the thesis. Therefore, time must be reserved for its elaboration;

11. Theses Defense: The final step in the program, one must defend the thesis one wrote.

4.2 Execution Schedule

This P.h.D. research proposal is schedule to start in August 2026 and run for the usual time for the Doctoral Program, that is, 48 months. The project will be run with weekly meetings with the supervisor, and sporadic seminar expositions to the other members of the research team and correlated groups such as the analysis and physical mathematics groups. We can divide the project executions in the following schedule based on the previous steps.

5 Methods and Material

This project follows the Materials and Methods usual to Mathematics: Extensive bibliographical research, courses in general and research-specific topics, participation in

Activity	Semester							
	1	2	3	4	5	6	7	8
1 - Bibliographical Research	*	*	*	*	*	*	*	*
2 - Guidance Meetings	*	*	*	*	*	*	*	*
3 - Program's Courses	*	*	*	*				
4 - Qualification Exam	*	*						
5 - FAPESP Technical Reports		*		*		*		*
6 - Research Problem Preliminary Study		*	*					
7 - Research Problem Solution Possibilities		*	*	*				
8 - Implementation of Possible Solutions				*	*	*	*	*
9 - Results Formulation and Publications						*	*	*
10 - Thesis Development						*	*	*
11 - Theses Defense								*

Tabela 1: Activities planned, the goals are as described in the Project. Red color indicates the focus of the quarter that may be divided between activities for some crucial quarters.

field-related events, research seminars and frequent meetings with the advisor. Discussion with peers from the program and the research groups is also expected to be a regular-basis occurrence. As much as possible, presentation about results and the general area are also very much encouraged for the gained exposition, experience and community. In the usual spirit of the scientific way, we will sought to extensively present all results in local, regional, national and international conferences and events. Furthermore, upon crystallization, results should be published to platforms of open access and suitable publishers.

6 Result Analysis

As is praxis for Mathematics the process of development of the results will be validated by weekly meetings with the supervisor and preliminary results will be presented in local seminars for discussion and possible community collaboration. As expected, final results will be presented in relevant conferences at the regional, national and possibly international conferences. Furthermore, preprint of publications will be posted to the ArXiv platform and subsequently submitted to high standard scientific journals.

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